

1. Introduction to Pandas

What is Pandas?

[Pandas Official Website](https://pandas.pydata.org/)

Pandas is a Python library used for:

- Data Analysis
- Data Cleaning
- Data Manipulation
- Working with Tables and Excel-like data

Pandas helps us work with:

- Rows
 - Columns
 - Tables
 - CSV files
 - Excel files
 - Database data
-

Real-Life Example

Suppose you have student data:

Name	Age	Marks
Rahul	20	85
Aman	21	90
Priya	19	88

Using Pandas, you can:

- Find average marks
- Filter students
- Sort data
- Remove empty values
- Analyze reports

Why Use Pandas in Data Analytics?

In Data Analytics, huge amounts of data are processed.

Without Pandas:

- Work becomes slow
- Manual calculations increase
- Data handling becomes difficult

With Pandas:

- Fast processing
- Easy analysis
- Less coding
- Better reporting

Example Without Pandas

```
marks = [85, 90, 88]

average = sum(marks) / len(marks)

print(average)
```

Output:

87.66

Example With Pandas

```
import pandas as pd

data = {
    "Marks": [85, 90, 88]
}

df = pd.DataFrame(data)

print(df["Marks"].mean())
```

Output:

87.66

Pandas makes calculations easier and faster.

Features of Pandas

1. Fast Data Processing

Pandas handles large datasets quickly.

Example:

```
df.sum()
```

2. Easy Data Cleaning

Remove missing or duplicate values easily.

Example:

```
df.dropna()
```

3. Works with CSV and Excel

Read files directly.

Example:

```
pd.read_csv("students.csv")
```

4. Powerful Filtering

Example:

```
df[df["Marks"] > 80]
```

5. Statistical Functions

Example:

```
df["Marks"].mean()  
df["Marks"].max()  
df["Marks"].min()
```

6. Data Visualization Support

Works with:

- [Matplotlib](#)
 - [Seaborn](#)
-

Installing Pandas

Before using Pandas, install it.

Step 1: Install Python

Download Python from:

[Python Official Website](#)

During installation:

☒ Check:

Add Python to PATH

Step 2: Open Command Prompt

Press:

Windows + R

Type:

cmd

Press Enter.

Step 3: Install Pandas

Run:

pip install pandas

Step 4: Verify Installation

Run:

pip show pandas

You will see:

- Version

- Location
 - Package details
-

Installing in Jupyter Notebook

If using [Jupyter Notebook](#):

```
!pip install pandas
```

Importing Pandas

After installation, import Pandas into Python.

Syntax

```
import pandas as pd
```

Meaning of This Line

Part	Meaning
import	Load library
pandas	Library name
as	Create shortcut
pd	Short name

Why Use "pd"?

Instead of writing:

```
pandas.DataFrame()
```

We write:

```
pd.DataFrame()
```

This saves time.

First Pandas Program

```
import pandas as pd

data = {
    "Name": ["Rahul", "Aman", "Priya"],
    "Marks": [85, 90, 88]
}

df = pd.DataFrame(data)

print(df)
```

Output

	Name	Marks
0	Rahul	85
1	Aman	90
2	Priya	88

Understanding the Code

Step 1

```
import pandas as pd
```

Imports Pandas.

Step 2

```
data = {  
    "Name": ["Rahul", "Aman", "Priya"],  
    "Marks": [85, 90, 88]  
}
```

Creates data in dictionary form.

Step 3

```
df = pd.DataFrame(data)
```

Converts dictionary into table format.

```
df means DataFrame.
```

Step 4

```
print(df)
```

Displays table.

What is DataFrame?

DataFrame is:

- Table format data
- Rows and columns
- Similar to Excel sheet

Example:

Name	Marks
Rahul	85
Aman	90

What is Series?

Series means:

- Single column data

Example:

```
marks = pd.Series([85, 90, 88])  
  
print(marks)
```

Output:

```
0    85  
1    90  
2    88
```

Summary

Topic	Explanation
Pandas	Python library for data analysis
DataFrame	Table data
Series	Single column
pd	Shortcut name
read_csv()	Read CSV file
mean()	Average
dropna()	Remove empty values